

Causal Inference for Infectious Disease Prevention

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November 17, 2025



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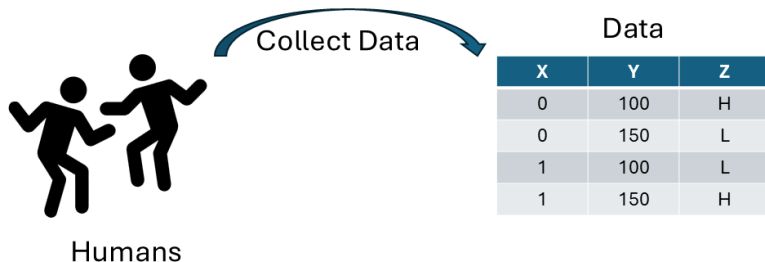


Why Biostatistics?

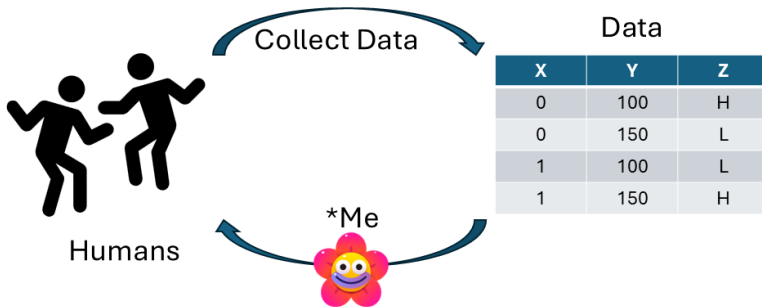


Humans

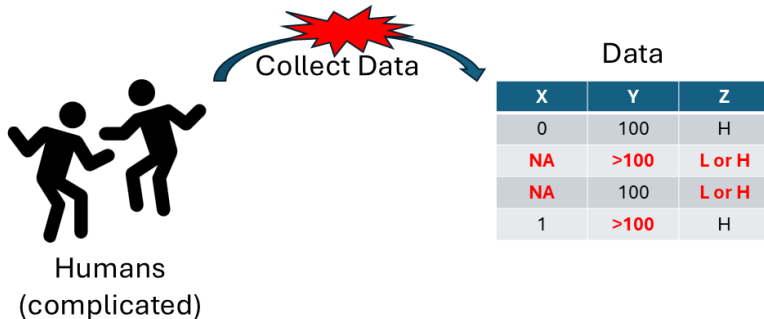
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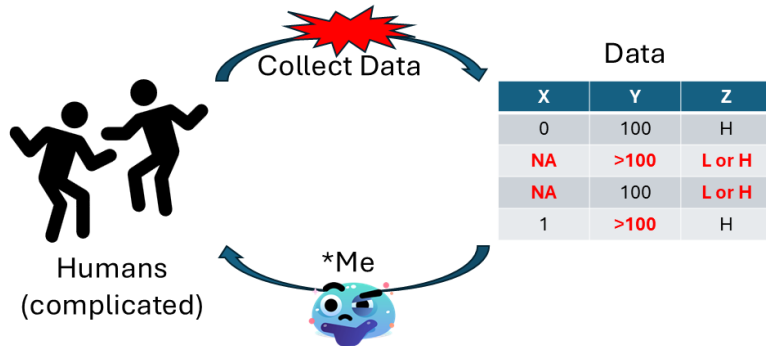
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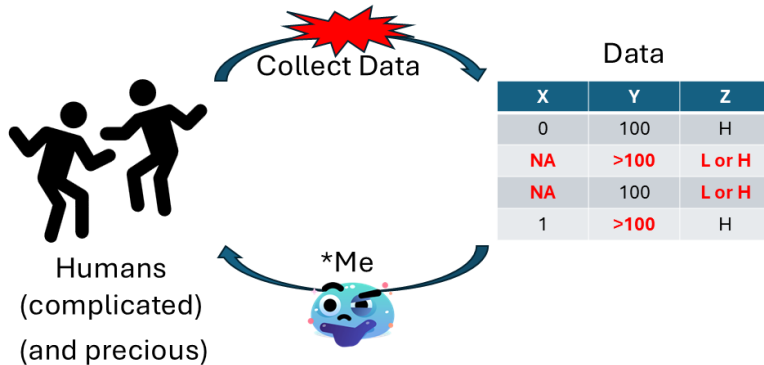
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Why Biostatistics?



Causal Inference

We are often interested in **causation**, as opposed to **association**

- **Associational question:** Do people who get the COVID-19 vaccine get sick?
- **Causal question:** If someone *were* to get the COVID-19 vaccine, *would* they get sick?

Associational Question



Associational Question

Vaccinated?



$$Z_i = 1$$



$$Z_j = 0$$

Associational Question

Vaccinated?



Sick?



Associational Question

Health-Conscious?



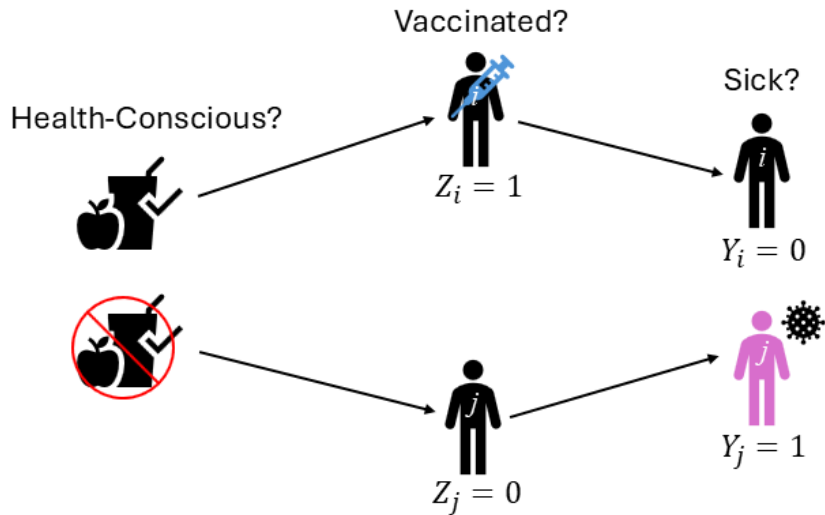
Vaccinated?



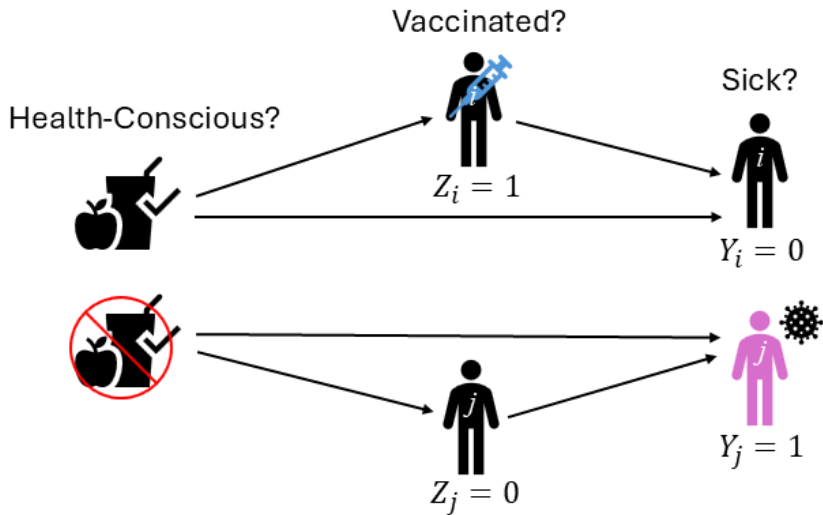
Sick?



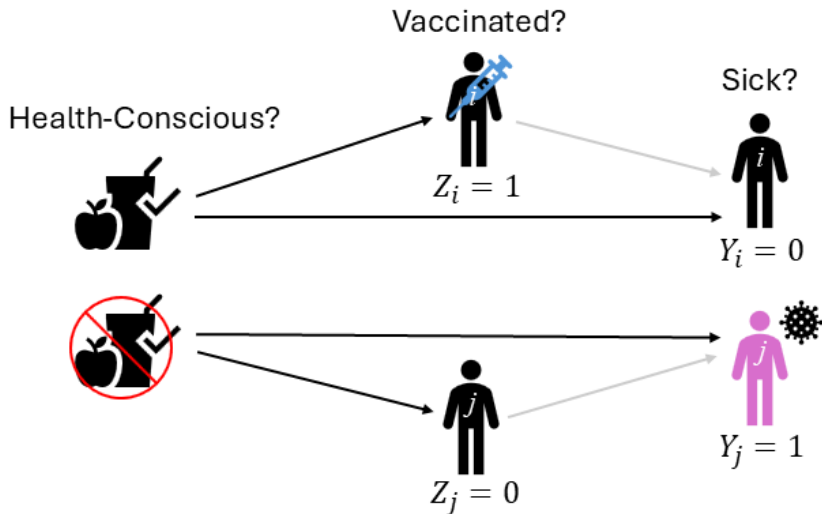
Associational Question



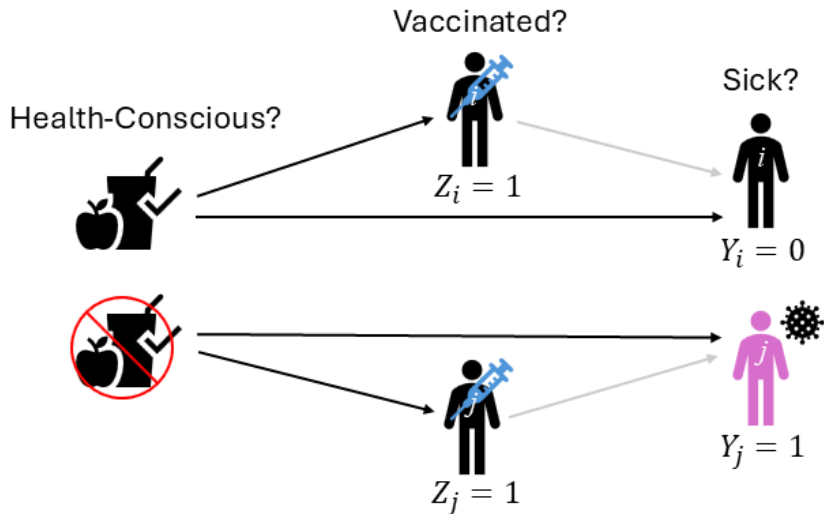
Associational Question



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Associational Question



My Causal Questions

- 1 If somebody were to have more antibody-dependent cellular phagocytosis, would this lower their risk of HIV infection?

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- ❶ If somebody were to have more antibody-dependent cellular phagocytosis, would this lower their risk of HIV infection?
- ❷ If better HIV prevention and treatment resources were available in a community, would this lower the incidence of HIV in that community?

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- ❸ If college students were told to social distance when sick with the flu, would this reduce transmission of the flu?

Social Distancing

If college students were told to social distance when sick with the flu, would this reduce transmission of the flu?

eX-FLU Trial

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season ([Aiello et al., 2016](#); [Zivich et al., 2020](#))

Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial

Allison E. Aiello^{a,*}, Amanda M. Simanek^{b,1}, Marisa C. Eisenberg^c, Alison R. Walsh^c, Brian Davis^c, Erik Volz^{d,1}, Caroline Cheng^c, Jeanette J. Rainey^e, Amra Uzicanin^e, Hongjiang Gao^e, Nathaniel Osgood^f, Dylan Knowles^f, Kevin Stanley^f, Kara Tarter^c, Arnold S. Monto^c

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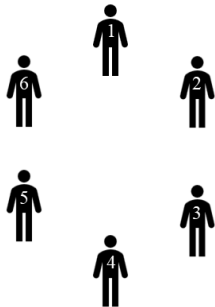
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- **Central question**: does the encouragement-to-isolate intervention reduce transmission of ILI?

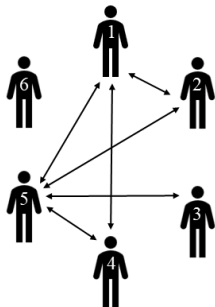
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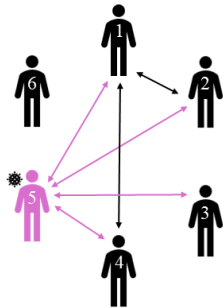
Transmission of Influenza-Like-Illness



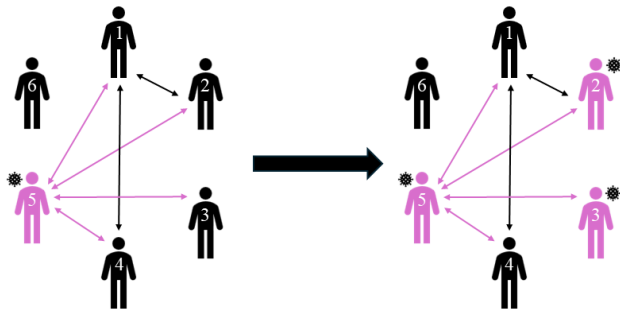
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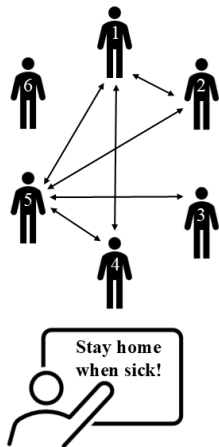
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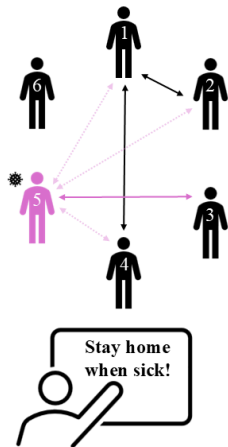
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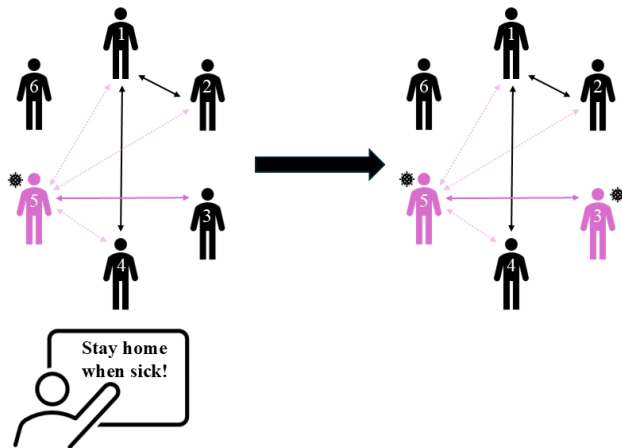
Example I: Intervention Affects Network and Transmission



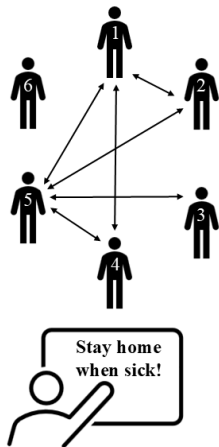
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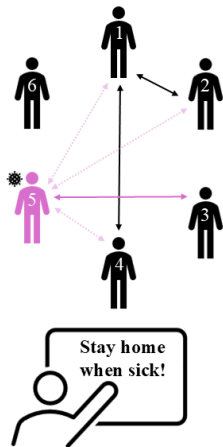
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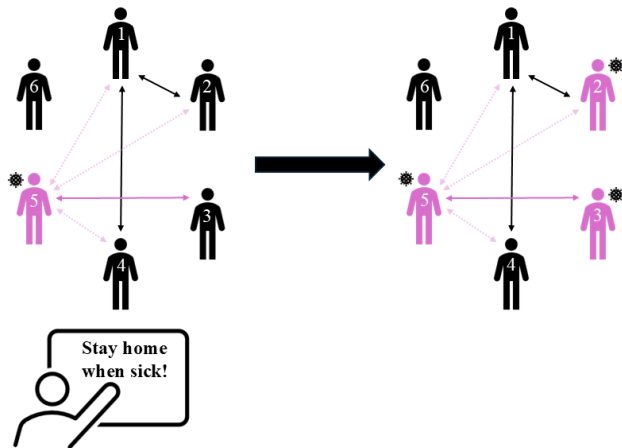
Example II: Intervention Affects Network, Not Transmission



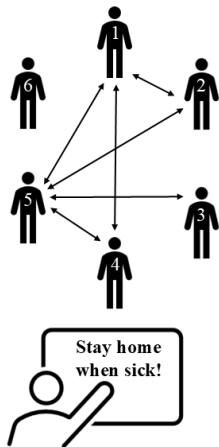
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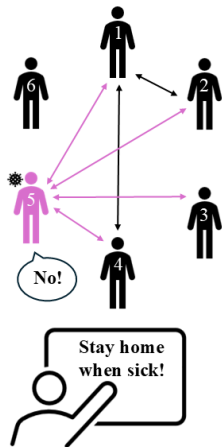
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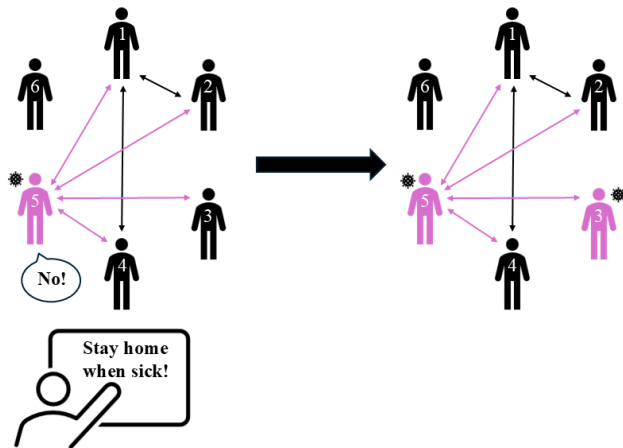
Example III: Intervention Affects Transmission, Not Network



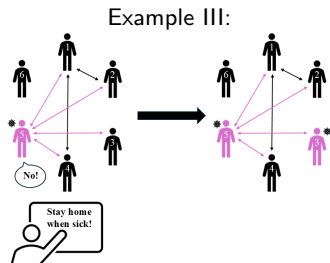
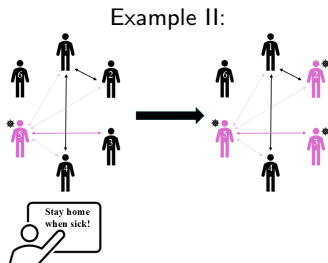
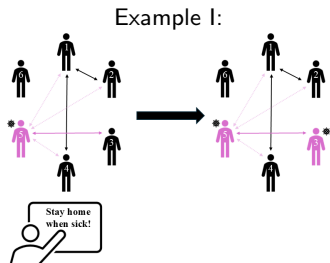
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Central Question: Does the Intervention Affect Transmission of ILI?



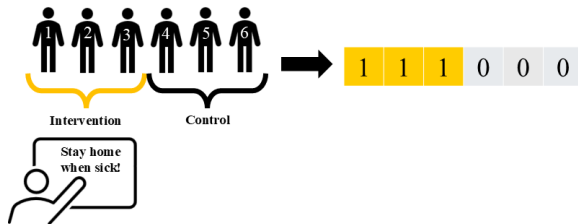
In Examples I and III, the answer is yes

eX-FLU Observed Data

- Baseline randomization assignments

$\mathbf{Z} = (Z_1, \dots, Z_n) \in \mathcal{Z}$, for

$Z_i = \mathbb{1}(\text{student } i \text{ gets intervention})$



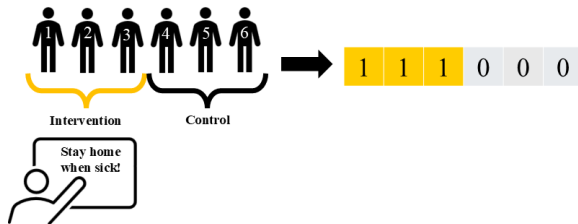
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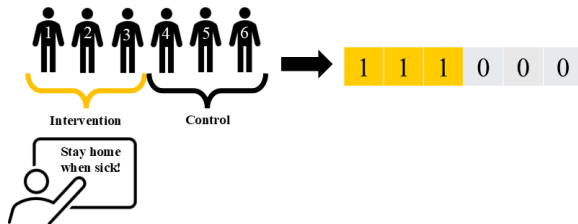
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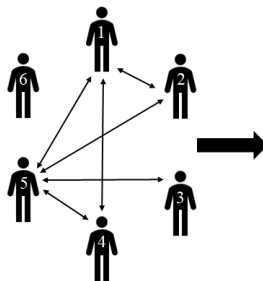
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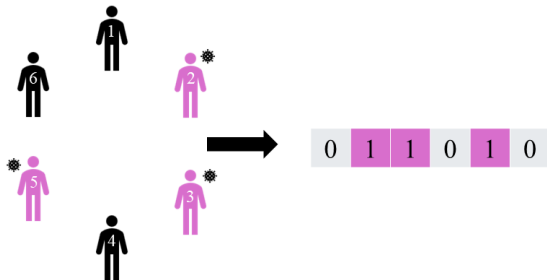
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	1	2	3	4	5	6
1	0	1	0	1	1	0
2	1	0	0	0	1	0
3	0	0	0	0	1	0
4	1	0	0	0	1	0
5	1	1	1	1	0	0
6	0	0	0	0	0	0

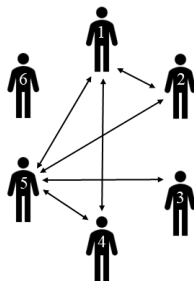
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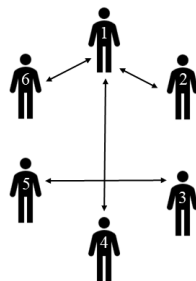
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Week 1

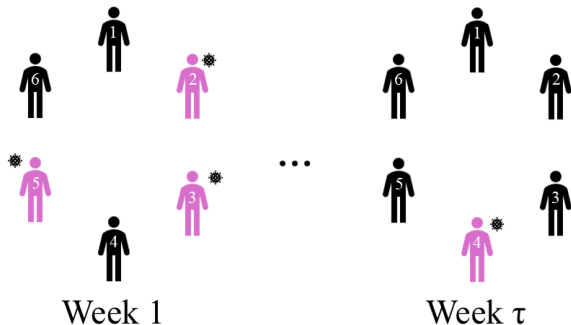
...



Week τ

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- network history: $\overline{\mathbf{A}} = \{\mathbf{A}^k\}_{k=1}^{\tau}$
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eX-FLU Potential Outcomes

Potential outcomes are defined for both the networks, and ILI infections:

$$\overline{\mathbf{A}}(\mathbf{z}), \quad \overline{\mathbf{Y}}(\mathbf{z}) \quad \text{for } \mathbf{z} \in \mathcal{Z}$$

- $\overline{\mathbf{A}}(\mathbf{z})$: sequence of networks that **would** occur if students **were** encouraged to isolate according to $\mathbf{z}$
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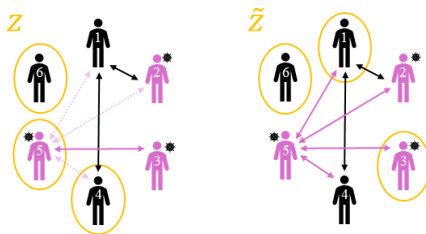
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- Assume **causal consistency**:

$$\overline{\mathbf{A}} = \overline{\mathbf{A}}(\mathbf{Z}) \text{ and } \overline{\mathbf{Y}} = \overline{\mathbf{Y}}(\mathbf{Z})$$

eX-FLU Null Hypotheses

$$H_0^Y : \bar{Y}(\mathbf{z}) = \bar{Y}(\tilde{\mathbf{z}}) \text{ for all } \mathbf{z}, \tilde{\mathbf{z}} \in \mathcal{Z}$$

(“the intervention has no effect on the infections”)



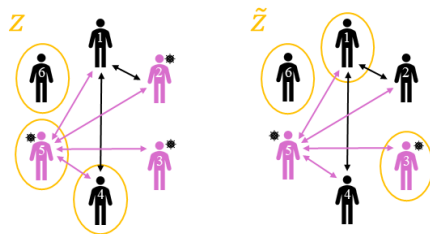
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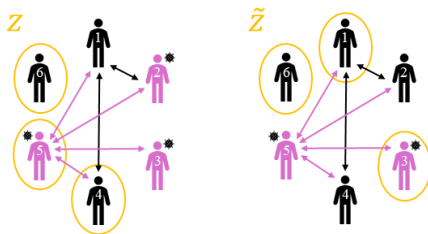
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$$H_0^\sharp = H_0^Y \cap H_0^A : \bar{Y}(\mathbf{z}) = \bar{Y}(\tilde{\mathbf{z}}) \text{ and } \bar{\mathbf{A}}(\mathbf{z}) = \bar{\mathbf{A}}(\tilde{\mathbf{z}}) \text{ for all } \mathbf{z}, \tilde{\mathbf{z}} \in \mathcal{Z}$$

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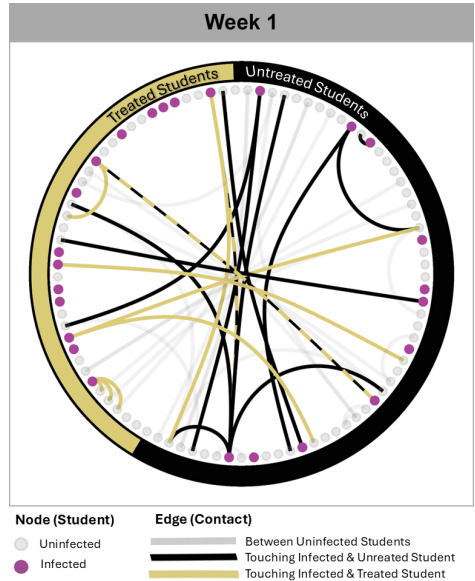
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Previous analyses of the eX-FLU trial ([Alexandria et al., 2023](#)) tested $H_0^\sharp$, but no analyses have tested H_0^Y

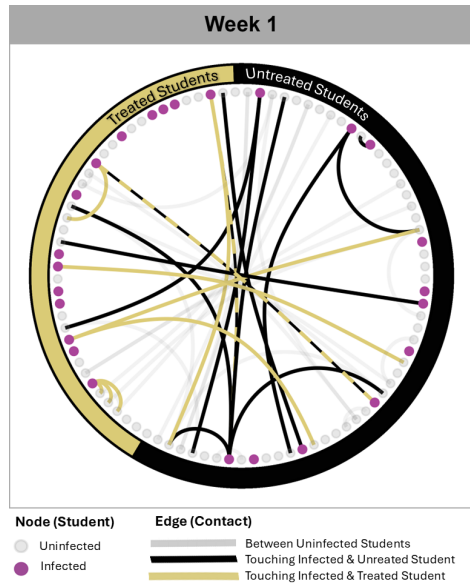
eX-FLU: Descriptive Results

- 93 (out of 579) students with at least one infection



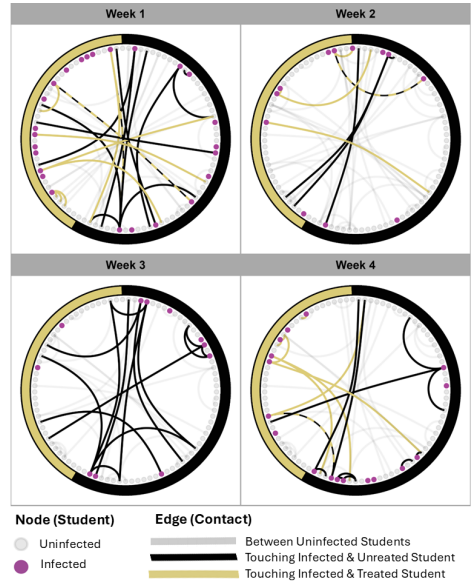
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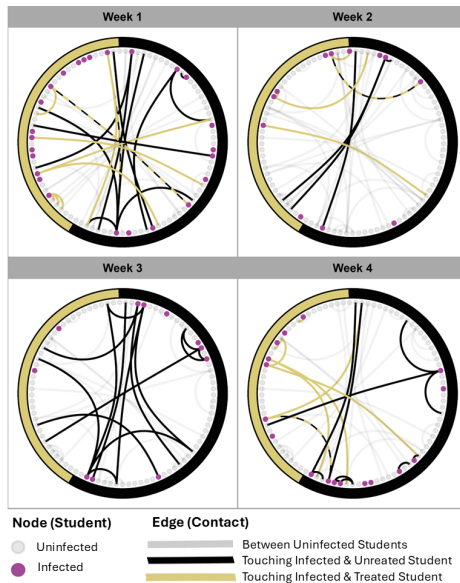
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eX-FLU: Descriptive Results

- 93 (out of 579) students with at least one infection
- Bold edges represent possible transmission events
 - Proportion of possible transmission events attributable to students in the intervention group:

$$\begin{aligned}
 T &= \frac{\text{number of yellow edges}}{\text{total number of edges}} = \\
 &= \frac{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Z_i Y_i^{k-1} A_{ij}^{k-1}}{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Y_i^{k-1} A_{ij}^{k-1}} \\
 &= 0.359
 \end{aligned}$$



Testing $H_0^\#$

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- Choose a test statistic $T = T(\mathbf{Z}, \overline{\mathbf{A}}, \overline{\mathbf{Y}})$

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- Find the distribution of T under $H_0^\sharp$

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Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$			
$\mathbf{z}_2$			
$\dots$			
$\mathbf{z}_{ \mathcal{Z} }$			

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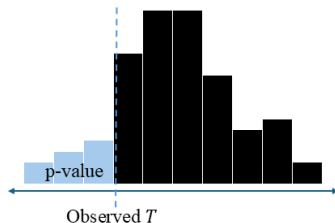
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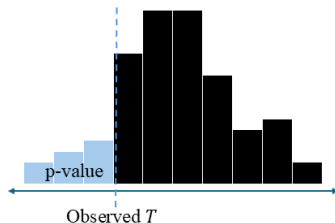
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- Reject $H_0^\sharp$ if $\rho^\sharp \leq 0.05$



Testing $H_0^\#$

- This test of $H_0^\#$ controls the **type I error rate** exactly:

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- ▶ In general, T should be chosen such that small values indicate departures from $H_0^\#$

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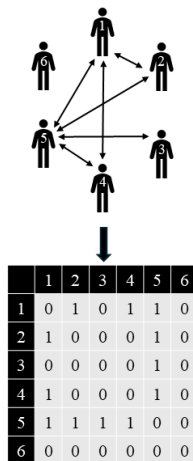
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 - ★ E.g., using a **Separable Temporal Exponential Random Graph Model**
 - ★ Use estimated q to obtain $\hat{\rho}^Y$

Exponential Family Random Graph Models

Goal: model a network ($\mathbf{A}$) given covariates ($\mathbf{X}$)

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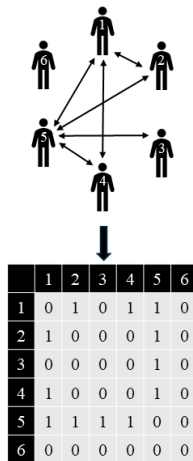


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An **ERGM** assumes:

$$\Pr_{\theta}(\mathbf{A} = \mathbf{a} | \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}}{\kappa(\theta, \mathcal{A}, \mathbf{x})}$$



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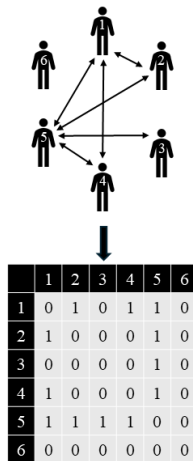
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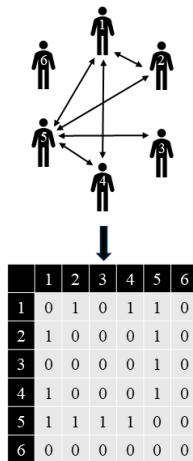
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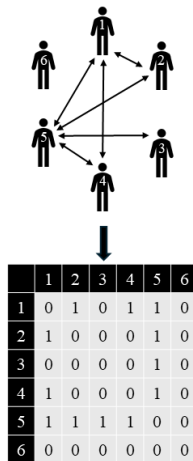
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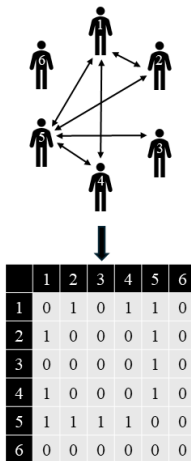
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$$\Pr_{\theta}(\mathbf{A} = \mathbf{a} | \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}}{\kappa(\theta, \mathcal{A}, \mathbf{x})}$$

- normalizing constant:

$$\kappa(\theta, \mathcal{A}, \mathbf{x}) = \sum_{\mathbf{a} \in \mathcal{A}} \exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}$$

- $\mathcal{A}$: space of possible networks
- θ : model coefficients
- $\mathbf{g}(\mathbf{a}, \mathbf{x})$: sufficient statistics



Exponential Family Random Graph Models

Goal: model a network (**A**) given covariates (**X**)

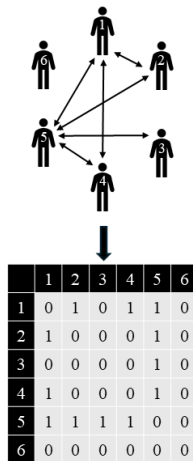
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 - ▶ number of edges



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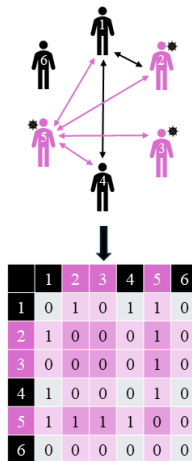
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 - ▶ number of edges
 - ▶ number of edges touching treated students



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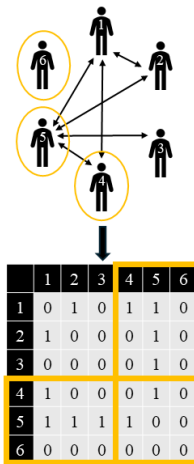
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 - ▶ number of edges touching infected students



Exponential Family Random Graph Models

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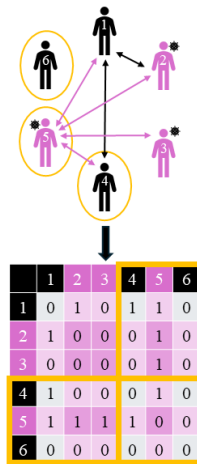
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 - ▶ number of edges
 - ▶ number of edges touching treated students
 - ▶ number of edges touching infected students
 - ▶ number of edges touching treated $\times$ infected students



ERGM Model Formulation

$$\Pr_{\theta}(\mathbf{A} = \mathbf{a} | \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}}{\kappa(\theta, \mathcal{A}, \mathbf{x})},$$

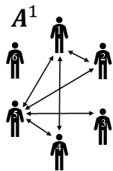
$$\kappa(\theta, \mathcal{A}, \mathbf{x}) = \sum_{\mathbf{a} \in \mathcal{A}} \exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}$$

For example, in the simulation study and eX-FLU application,

$$\mathbf{g}(\mathbf{a}, \mathbf{x}) = \begin{bmatrix} \# \text{ edges} \\ \# \text{ edges touching a treated node} \\ \# \text{ edges touching an infected node} \\ \# \text{ edges touching a treated and infected node} \\ \# \text{ edges between roommate pairs} \end{bmatrix} = \begin{bmatrix} \sum_{i,j} a_{ij} \\ \sum_{i,j} a_{ij}(z_i + z_j) \\ \sum_{i,j} a_{ij}(y_i + y_j) \\ \sum_{i,j} a_{ij}(z_i y_i + z_j y_j) \\ \sum_{i,j} a_{ij} \mathbb{1}(i, j \text{ roommates}) \end{bmatrix}$$

Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

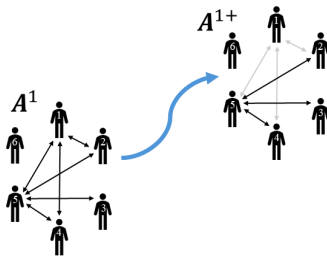


Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

- New edges form according to a **formation** ERGM

$$\Pr_{\theta^+}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^+ \cdot \mathbf{g}^+(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{x}\}}$$

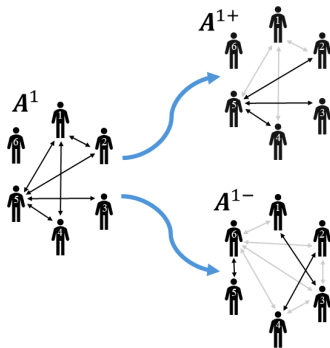


Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

- New edges form according to a **formation** ERGM
- Old edges persist according to a **persistence** ERGM

$$\Pr(\mathbf{A}^{k-} = \mathbf{a}^{k-} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^- \cdot \mathbf{g}^-(\mathbf{a}^{k-}, \mathbf{x})\}}{\kappa\{\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{x}\}}.$$

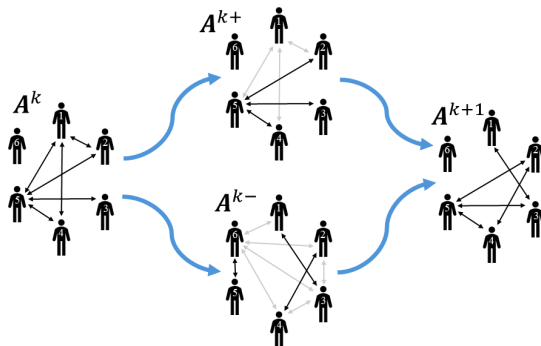


Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

- New edges form according to a **formation** ERGM
- Old edges persist according to a **persistence** ERGM
- The network at time step $k + 1$ is the result of formation and persistence

$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^k}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+} - \mathbf{A}^k)}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^k - \mathbf{A}^{k-})}_{\text{old edges not persisting}}$$



Testing H_0^Y

Testing Procedure:

- 1 Estimate the unknown parameter θ in the PMF $q(\bar{\mathbf{a}}, \mathbf{z}; \theta)$ of $\bar{\mathbf{A}}(\mathbf{z})$

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for $b = 1$ to B do:

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for $b = 1$ to B **do**:
 - ① generate randomization vector $\mathbf{z}_b$

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- ③ compute the plug-in p-value testing H_0^Y :

$$\hat{\rho}_B^Y = B^{-1} \sum_{b=1}^B \mathbb{1}(T_b \leq T)$$

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Proposition: This procedure will **asymptotically** control the type I error rate

Simulation Study

- $n \in \{24, 48\}$ students
- $\tau \in \{5, 10\}$ weeks
- 50:50 cluster randomization with 12 equal-sized clusters

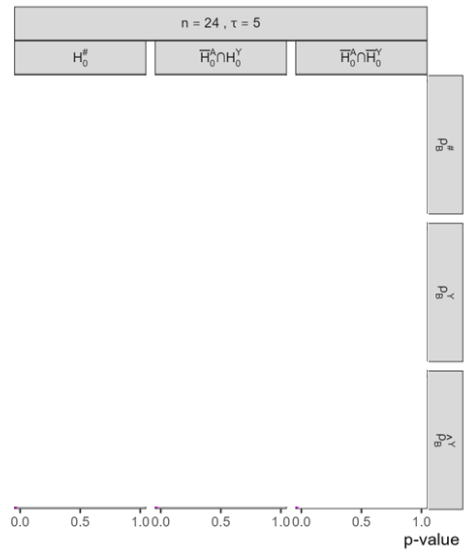
Simulation Study

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- Three scenarios:
 - 1 $H_0^\sharp$: no effect of intervention on networks or infection
 - 2 $\overline{H}_0^A \cap H_0^Y$: no effect of intervention on infection
 - 3 $\overline{H}_0^A \cap \overline{H}_0^Y$: effect of intervention on both networks and infection

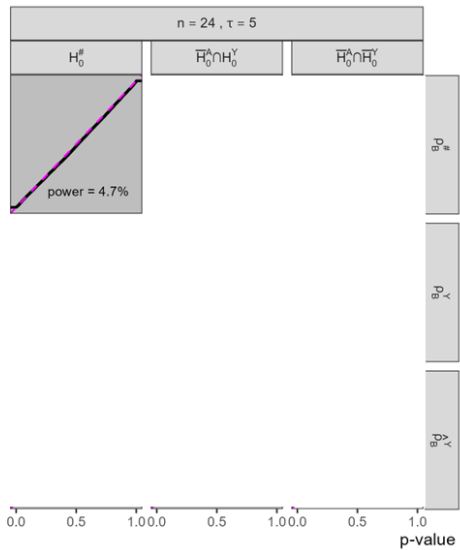
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- Three testing procedures:
 - 1 $\rho_B^\#$: testing $H_0^\#$
 - 2 ρ_B^Y : testing H_0^Y using known q
 - 3 $\hat{\rho}_B^Y$: testing H_0^Y using estimated q

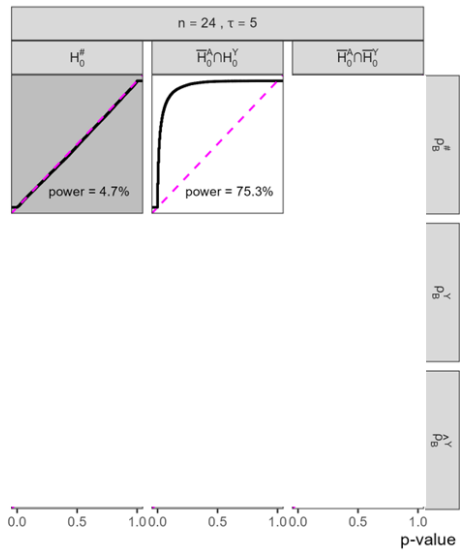
Simulation Results



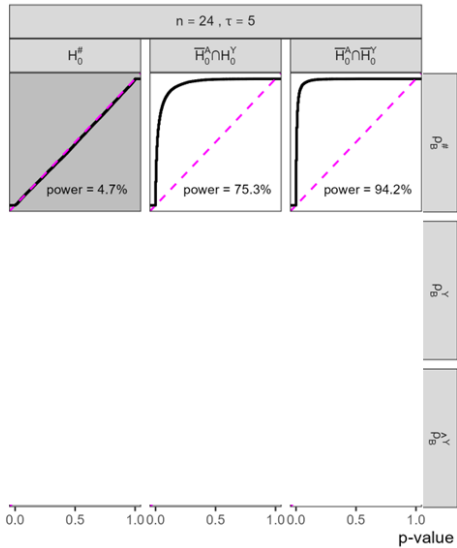
Simulation Results



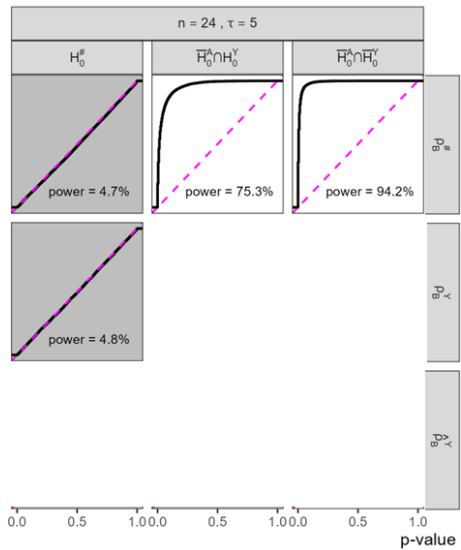
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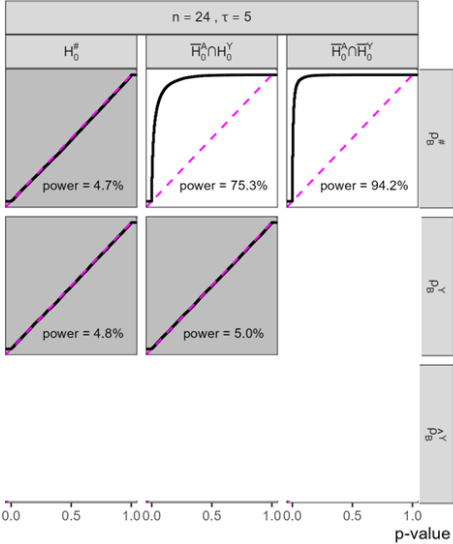
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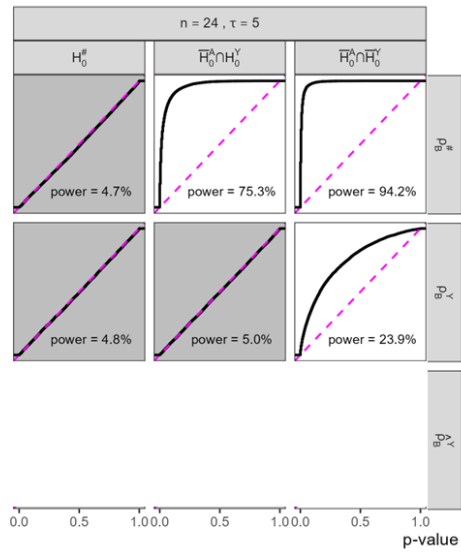
Simulation Results



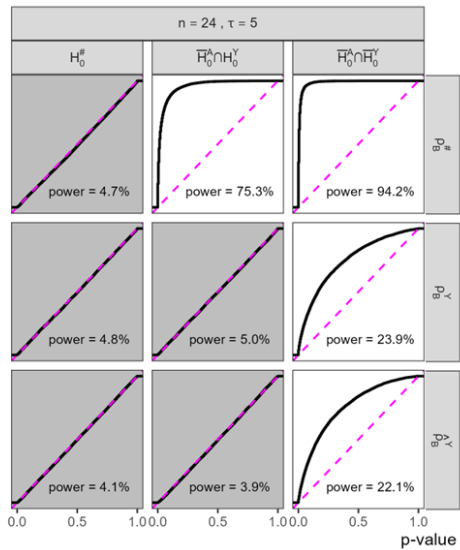
Simulation Results



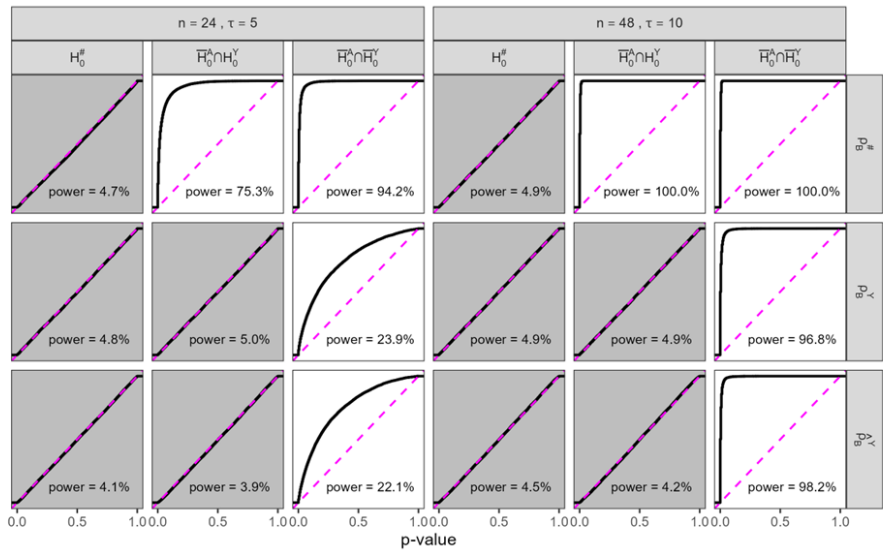
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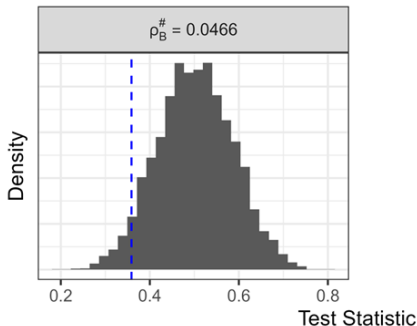


eX-FLU: Hypothesis Test Results

- Test statistic (proportion of possible transmission events attributable to students in the intervention group):
 $T = 0.359$.

eX-FLU: Hypothesis Test Results

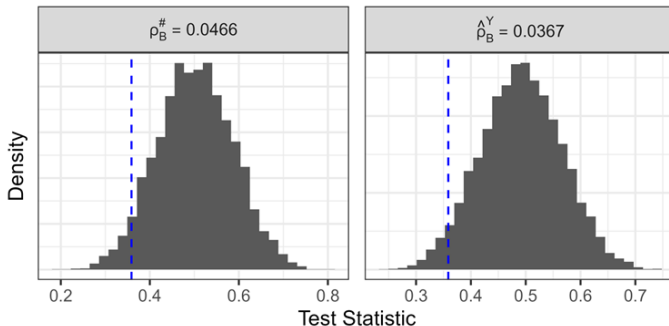
- Test statistic (proportion of possible transmission events attributable to students in the intervention group):
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- Encouragement to isolate affects the social network and/or influenza-like illness ($\rho_B^\# = 0.0466$)

eX-FLU: Hypothesis Test Results

- Test statistic (proportion of possible transmission events attributable to students in the intervention group):
 $T = 0.359$.



- Encouragement to isolate affects the social network and/or influenza-like illness ($\rho_B^\# = 0.0466$)
- Encouragement to isolate specifically affects influenza-like illness ($\hat{\rho}_B^Y = 0.0367$)

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Choice of Test Statistic

Test statistics: “proportion of possible transmission events attributable to students in the intervention group:”

$$T^{**}(\mathbf{Z}, \bar{\mathbf{A}}, \bar{\mathbf{Y}}) = \frac{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Z_i E_{ijk}^{**}}{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} E_{ijk}^{**}}.$$

	From Infected to Infected	From Infected
Contact at $k - 1$	$E_{ijk}^{11} = Y_i^{k-1} A_{ij}^{k-1} Y_j^k$	$E_{ijk}^{12} = Y_i^{k-1} A_{ij}^{k-1}$
Contact at k	$E_{ijk}^{21} = Y_i^{k-1} A_{ij}^k Y_j^k$	$E_{ijk}^{22} = Y_i^{k-1} A_{ij}^k$
Contact at $k - 1$ or k	$E_{ijk}^{31} = Y_i^{k-1} (A_{ij}^{k-1} \vee A_{ij}^k) Y_j^k$	$E_{ijk}^{32} = Y_i^{k-1} (A_{ij}^{k-1} \vee A_{ij}^k)$
Contact at $k - 1$ and k	$E_{ijk}^{41} = Y_i^{k-1} (A_{ij}^{k-1} * A_{ij}^k) Y_j^k$	$E_{ijk}^{42} = Y_i^{k-1} (A_{ij}^{k-1} * A_{ij}^k)$

Table: Definitions of a possible transmission event E_{ijk} from student i to student j at time k . Y_i^k is an indicator for student i being infected at week k , A_{ij}^k is an indicator for students i and j being in contact at week k , and $a \vee b$ denotes the maximum of a and b .

ERGM Change Statistic Model Formulation

- **Change statistic:** $\delta_{\mathbf{g}}(\mathbf{a}, \mathbf{x})_{ij} = \mathbf{g}(\mathbf{a}_{ij}^+, \mathbf{x}) - \mathbf{g}(\mathbf{a}_{ij}^-, \mathbf{x})$ is the change in network statistic that would occur if a_{ij} were changed from 0 to 1
 - ▶ where $\mathbf{a}_{ij}^+$ and $\mathbf{a}_{ij}^-$ represent the network $\mathbf{a}$ with dyad a_{ij} set to 1 or 0, respectively
- Then the equivalent ERGM specification is

$$\text{logit}\{\Pr(A_{ij} = 1 | \mathbf{A}_{ij}^C = \mathbf{a}_{ij}^C, \mathbf{X} = \mathbf{x})\} = \theta^k \delta_{\mathbf{g}}(\mathbf{a}, \mathbf{x})_{ij}$$

- ▶ where $\mathbf{A}_{ij}^C$ represents all dyads in $\mathbf{A}$ except A_{ij}
- **Interpretation of θ :** the change in conditional log-odds of the network associated with a one-unit increase in the corresponding component of $\mathbf{g}(\mathbf{a}, \mathbf{x})$ resulting from switching a particular dyad A_{ij} from 0 to 1 and leaving the rest of the network fixed at $\mathbf{A}_{ij}^C$

Dyadic Independence EGRMs

- **Dyadic independence term:** a component g of $\mathbf{g}$ in an ERGM for which the corresponding change statistic $\delta_g(\mathbf{a}, \mathbf{x})_{ij}$ can be calculated for any i, j without knowing $\mathbf{a}$
 - ▶ For example, if $g(\mathbf{a}, \mathbf{x}) = \sum_{i,j} a_{ij}(Z_i + Z_j)$ counts the number of edges touching treated nodes, then $\delta_g(\mathbf{a}, \mathbf{x})_{ij} = z_i + z_j$ doesn't depend on $\mathbf{a}$
- **Dyadic independence ERGM:** an ERGM with only dyadic independence terms
 - ▶ replace $\delta_g(\mathbf{a}, \mathbf{x})_{ij}$ with $\delta_g(\mathbf{x})_{ij}$ and write the model as

$$\text{logit}\{\Pr(A_{ij} = 1 | \mathbf{X} = \mathbf{x})\} = \theta \cdot \delta_g(\mathbf{x})_{ij}$$

- **Interpretation of θ :** the change in log-odds of the network associated with a one-unit increase in the corresponding component of $\mathbf{g}(\mathbf{a}, \mathbf{x})$ resulting from switching a particular dyad A_{ij} from 0 to 1

STERGM Model Formulation

- **Formation model** is an ERGM conditional on only adding edges:

$$\Pr_{\theta^+}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^+ \cdot \mathbf{g}^+(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{x}\}}$$

► $\mathcal{A}^+(\mathbf{a})$: space of possible networks that can be formed by adding edges to $\mathbf{a}$

- **Persistence model** is an ERGM conditional on only removing edges:

$$\Pr_{\theta^-}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^- \cdot \mathbf{g}^-(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{x}\}}$$

► $\mathcal{A}^-(\mathbf{a})$: space of possible networks that can be formed by removing edges to $\mathbf{a}$

- A **STERGM** assumes the network at time $k + 1$ is then the result of applying the changes in $\mathbf{A}^{k+}$ and $\mathbf{A}^{k-}$ to $\mathbf{A}^k$:

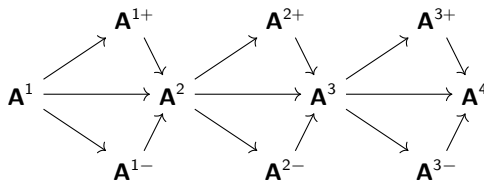
$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^k}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+} - \mathbf{A}^k)}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^k - \mathbf{A}^{k-})}_{\text{old edges not persisting}}$$

Separability of STERGMs

- 1 $\mathbf{A}^{k+} \perp\!\!\!\perp \mathbf{A}^{k-} | \mathbf{A}^k$, i.e., the formation and persistence processes are conditionally independent given the network at time k

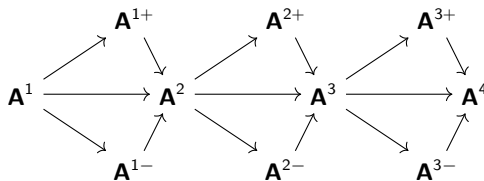
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Separability of STERGMs

- 1 $\mathbf{A}^{k+} \perp\!\!\!\perp \mathbf{A}^{k-} | \mathbf{A}^k$, i.e., the formation and persistence processes are conditionally independent given the network at time k
- 2 the parameter space for $\theta = (\theta^+, \theta^-)$ is the product of the parameter spaces for θ^+ and θ^-



Simulation Setup

- $n \in \{24, 48\}$ students were equally divided into two residence halls, each with six clusters of equal size
- Within each residence hall group, the clusters were randomized using a 50:50 allocation to either the intervention group or the control group
- Five pairs of students were randomly selected to be roommates, meaning they had contact with each other each week
- Baseline ($k = 1$) social contacts were simulated between each pair of students with probability 0.5
- Baseline infection statuses were simulated for each student with probability 0.5
- Social networks were simulated over the remaining $\tau \in \{5, 10\}$ weeks according to a STERGM with both formation and persistence models including edge count, intervention assignment Z , infection status Y , $Z \times Y$ interaction, and an offset to force constant edges between roommates
- Infection statuses were simulated for each student i at each week $k \in \{2, \dots, \tau\}$ with probability $\Pr(Y_i^k = 1 | \mathbf{A}^{k-1}, \mathbf{Y}^{k-1}) = g(\sum_{j=1}^n A_{ij}^{k-1} Y_j^{k-1})$, where $\sum_{j=1}^n A_{ij}^{k-1} Y_j^{k-1}$ is the number of infected contacts at the previous week, and $g : [0, \infty) \rightarrow [0, 1]$ is a non-decreasing function

Simulation Setup

Three scenarios:

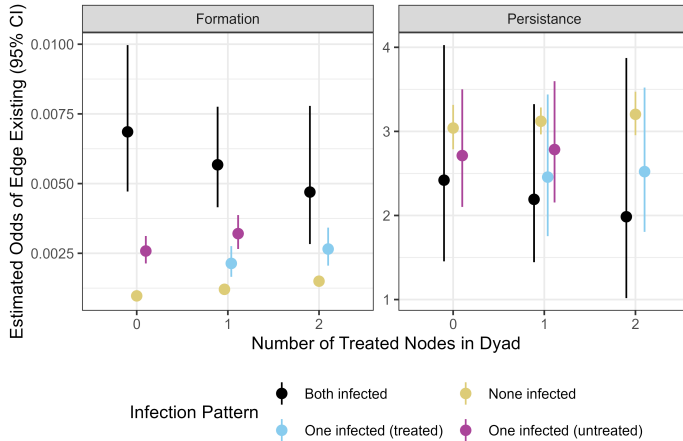
Null Hypothesis	Formation Model Parameters θ^+	Persistence Model Parameters θ^-	Infection Probability Function g
$H_0^\sharp$	$(-0.5, 0, 0, 0)$	$(-0.5, 0, 0, 0)$	$g(s) = 0.5$
$\overline{H}_0^A \cap H_0^Y$	$(-0.2, 0, 0, -1)$	$(-0.2, 0, 0, -1)$	$g(s) = 0.5$
$\overline{H}_0^A \cap \overline{H}_0^Y$	$(-0.2, 0, 0, -1)$	$(-0.2, 0, 0, -1)$	$g(s)$ increasing in s

Table: Data generating process for the simulation study. The null hypothesis refers to the null hypothesis that is true under the data generating process, formation model parameters are $\theta^+ = (\theta_{\text{edges}}^+, \theta_Z^+, \theta_Y^+, \theta_{ZY}^+)$, persistence model parameters are $\theta^- = (\theta_{\text{edges}}^-, \theta_Z^-, \theta_Y^-, \theta_{ZY}^-)$, and the infection probability function g provides a unit's probability of being infected at week k given their number of infected neighbors at week $k - 1$.

Three p-values:

- $\rho_B^\sharp$: testing the sharp null
- ρ_B^Y : testing H_0^Y using known q
- $\hat{\rho}_B^Y$: testing H_0^Y using estimated q

eX-FLU: STERGM Results



Type I Error Control

- **Proposition 2.1:** Let T_N be a test statistic with CDF F_N under H_0 .
 - ① Under H_0 , $F_N(T_N)$ stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ② If the test statistic T_N has a continuous limiting distribution, then $F_N(T_N) \rightarrow^d \text{Uniform}(0, 1)$.
- **Corollary 2.2:**
 - ① Under $H_0^\sharp$, the sharp null p-value $\rho_N^\sharp$ stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ② Under H_0^Y , the oracle p-value ρ_N^Y stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ③ If the test statistic T_N has a continuous limiting distribution, then $\rho_N^\sharp \rightarrow^d \text{Uniform}(0, 1)$ under $H_0^\sharp$ and $\rho_N^Y \rightarrow^d \text{Uniform}(0, 1)$ under H_0^Y .

Type I Error Control

- **Proposition 2.3:** Let $q(\mathbf{a}, \mathbf{z}, \theta) \equiv \Pr\{\mathbf{A}(\mathbf{z}) = \mathbf{a}; \theta\}$ denote the PMF of the distribution of stochastic potential networks $\mathbf{A}(\mathbf{z})$ at parameter value θ , let $\hat{\theta}_N$ denote the estimator of θ , and let $F_N(\cdot; \theta)$ denote the CDF of the test statistic T_N at θ , with limiting CDF $F(\cdot; \theta)$. Let θ_0 denote the true value of θ . Assume the following:

(A1) $\hat{\theta}_N \rightarrow^P \theta_0$

(A2) $F(t; \theta_0)$ is continuous in t on $\mathbb{R}$

(A3) there exists a $\delta_0 > 0$ such that

$$\sup_{\theta \in B_{\delta_0}(\theta_0)} \sup_{t \in \mathbb{R}} |F_N(t; \theta) - F(t; \theta)| \rightarrow 0$$

(A4) $F(t; \theta)$ is continuous in θ at θ_0 uniformly in t , i.e.,

$$\lim_{\theta \rightarrow \theta_0} \sup_{t \in \mathbb{R}} |F(t; \theta) - F(t; \theta_0)| = 0$$

Then the plug-in p-value $\hat{\rho}_N^Y$ converges in distribution to $\text{Uniform}(0, 1)$.

Type I Error Control

- **Proposition 2.4:** Let $\rho_N = F_N(T_N)$ for test statistic T_N and CDF F_N (not necessarily the true CDF of T_N). Let $T_N^* = h_N(T_N)$ for a sequence of deterministic, strictly increasing functions h_N . Define $F_N^*(t) = F_N\{h_N^{-1}(t)\}$, the (not necessarily true) CDF of the transformed test statistic, and let $\rho_N^* = F_N^*(T_N^*)$. Then $\rho_N^* = \rho_N$.
- **Corollary 2.5:** Let h_N be a sequence of deterministic, strictly increasing functions.
 - 1 If the hypotheses of Proposition 2.1 are met for a test statistic T_N , then the results also hold for $T_N^* = h_N(T_N)$.
 - 2 If the hypotheses of Proposition 2.3 are met for $T_N, F_N(\cdot; \theta)$, then the results also hold for $T_N^* = h_N(T_N), F_N^*(\cdot; \theta) = F_N\{h_N^{-1}(\cdot); \theta\}$.